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Course: Computer Vision

I. Smoothing using mean filter:

The mean filter is used to reduce random noise in an image. It works by replacing the value of its neighbouring pixels.

The filter uses a small mask such as 3×3 or 5×5 which moves across the image. For every pixel, the average of all pixel values inside the mask is calculated and assigned to the center pixel. This process smooths sudden intensity changes caused by random noise.

Expected outcome:

- Reduces random intensity variations
- Produces a smoother image
- Slightly blurs edges and fine details.
- Suitable for basic image preprocessing

2. Threshold based segmentation:

- The most suitable segmentation technique is thresholding. It separates the object from the background based on pixel intensity values.
- A threshold is selected. Every pixel with an intensity greater than the threshold is classified as the object, while pixels with intensity less than the threshold are classified as the background.
- This method is highly effective when the object is much brighter than its background because there is a clear difference in intensity values.

Advantages:

- Easy to implement
- Fast processing
- Works well when object and background have different brightness levels.

Expected outcome:

- The brighter object is cleanly extracted from the darker background, making further image analysis easier.

3. Sobel edge detection:

- It is a method widely used to identify objects in an image. It detects edges by measuring changes in pixel intensity in both horizontal & vertical directions.
- The sobel operator uses two 3×3 convolution masks. One mask detects horizontal edges while other detects vertical edges. The gradient values obtained both masks are combined to determine the strength and direction.

of the edges.

- The sobel method is simple, reliable and less sensitive to small amount of noise compared to basic gradient operators. It highlights object boundaries clearly while preserving the overall image structure.

- Advantages:

- Detects horizontal and vertical edges effectively
- Simple to implement
- Provides clear edge information.

- Expected outcome:

The image displays sharp and well defined object boundaries that can be used for segmentation and feature extraction.